

Tutorial No. 1

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Problem Statement: Analyze the causes and consequences of the software crisis and explain how software engineering principles address these issues.

Solution:

Causes of the software crisis

The software crisis refers to the difficulties faced in developing reliable, large-scale software systems during the early days of computing.

The major causes are:

Increasing software complexity - Growth in size and complexity made software difficult to manage.

Poor project management:- Lack of proper planning and unrealistic schedules.

Inadequate requirement analysis - Unclear and changing user requirements.

Lack of standard development methods - Absence of structured development methods.

Limited skilled manpower - Shortage of trained software professionals.

Poor quality assurance - Insufficient testing and validation.

Consequences of the software crisis

The above causes resulted in several negative consequences:

Project delays - Failure to complete projects on time

Cost overruns - Budgets exceeded due to rework

Low quality software - Frequent failures and bugs.

User dissatisfaction - Software failed to meet user needs

High maintenance cost - difficult and expensive maintenance

Project failures - Many projects were abandoned

Role of software engineering principles

Software engineering provides systematic approaches to overcome the software crisis:

SDLC models - Structured models like waterfall and agile improve control.

Requirement engineering - Proper requirement analysis and documentation.

Modularity and Abstraction - Reduces complexity

Software design principles - Improves maintainability and scalability

Quality Assurance and Testing - early defect detection.

Project management techniques - cost, time, and risk control.

Standards and documentation - Improves consistency

Conclusion:

The software crisis highlighted the need for disciplined software development practices. Software engineering principles help deliver reliable, maintainable, and cost-effective software systems.